Lignocaine 1%

Mechanism of action

- Lignocaine is a local anaesthetic.
- Lignocaine blocks the initiation and transmission of nerve impulses by blocking the movement of sodium ions across the nerve cell membrane.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Subcutaneous injection for prophylaxis of pain associated with IV cannulation.
- ✓ Subcutaneous injection for digital ring blocks for analgesia.
- ✓ Intraosseous injection for significant bone pain associated with intraosseous administration.

Contraindications

- ✓ Known severe allergy.
- ✓ Local infection in the area of injection.

Cautions

- ⚠ Taking an anticoagulant (ring blocks).

Use in pregnancy or when breastfeeding

- Safe. May be administered if indicated.

Dosage

- Subcutaneous:
  - The maximum subcutaneous dose for an adult is 200 mg (20 ml of 1% lignocaine).
  - See the paediatric drug dose tables for the maximum dose for a child.

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- The subcutaneous dose may be repeated once after 30 minutes.

- Intraosseous:
  - 50 mg (5 ml of 1% lignocaine) for an adult.
  - See the paediatric drug dose tables for a child.
  - The intraosseous dose may be repeated once after 15 minutes.

Administration

- Subcutaneous for IV insertion: administer into the subcutaneous tissue at the site of cannulation. Raise a bleb and wait approximately one minute before insertion.
- Digital ring blocks: administer approximately 1-2 ml of 1% lignocaine into the tissue on either side of the web space of the digit.
- Intraosseous: administer slowly over 1-2 minutes and wait one further minute before infusing fluid. This is intended to limit the amount of lignocaine flushed into the circulation.

Common adverse effects

- Stinging at the time of injection.

Usual onset of effect

- 1-2 minutes for IV cannulation.
- 5-10 minutes for digital ring blocks.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 50 mg in 5 ml.

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Pharmacokinetics

- Lignocaine is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Additional information

- Do not apply lignocaine topically to the eye because the solution contains a preservative that may cause harm.
- Warming lignocaine, for example in your pocket or hand, may reduce the stinging associated with subcutaneous injection.
- Overdose of lignocaine when administered subcutaneously is very rare, but can occur if doses exceed 3 mg/kg or more than 1 mg/kg is inadvertently administered IV. If this occurs the following may develop:
  - Tingling around the mouth.
  - Seizures.
  - Dysrhythmias, particularly bradydysrhythmias.
  - Hypotension.
  - Cardiac arrest.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications updated.

22 November 2022 1.0.1
Update: Aligned with analgesia CPG.